

Approximation algorithms: Linear and Integer Programming

Maria Serna

Spring 2025-2026

- 1 LP and IP
- 2 Relax and round
- 3 LP Duality

Linear programming

- In a linear programming problem, we are given a **set of variables**, an **objective linear function** a set of **linear constraints** and want to assign real values to the variables as to:
 - satisfy the set of linear inequalities (equations or constraints),
 - maximize or minimize the objective function.
- LP is a pure algebraic problem.

Linear programming: An example

$$\begin{aligned} & \max(x_1 + 6x_2) \\ & \text{subject to} \\ & \quad x_1 \leq 200 \\ & \quad x_2 \leq 300 \\ & \quad x_1 + x_2 \leq 400 \\ & \quad x_1, x_2 \geq 0 \end{aligned}$$

Linear programming: feasible region

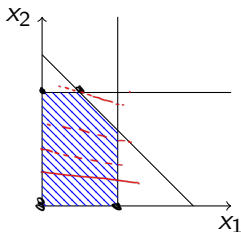
- A linear equality defines a **hyperplane**.
- A linear inequality defines a **half-space**.
- The solutions to the linear constraints lie inside a **feasible region** limited by the polytope (convex polygon in $\mathbb{R}^2$) defined by the linear constraints.

• vertices que delimiten
la solución!

— función objetivo

Si encontramos el óptimo,

ya no necesitamos buscar el otro vértice porque la solución



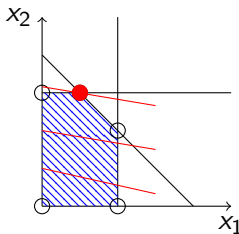
Linear programming: infeasibility

- A linear programming is **infeasible** if

- (mult
estimate) • The constraints are so tight that it is impossible to satisfy all of them.
 For ex. $x \geq 2$ and $x \leq 1$
- (no
estimate) • The constraints are so loose that the feasible region is unbounded
 allowing the objective function to go to ∞ .
 For ex. $\max x_1 + x_2$ subject to $x_1, x_2 \geq 0$

Linear programming: optimum

- In a feasible linear programming the **optimum** is achieved at a **vertex** of the feasible region.



Linear programming: standard formulation

A LP has many degrees of freedom.

- maximization or minimization.
 - constraints could be $=$, $\geq$, $\leq$, $<$ or $>$.
 - variables are often restricted to be non-negative, but they also could be unrestricted.
-
- standard form?

Linear programming: standard formulation

- From max to min (or min to max)
multiply by -1 the coefficients of the objective function.
- To reverse an inequality (for ex. $\geq$ to $\leq$)
multiply all coefficients and the independent term by -1.
- From $<$ to $\leq$ (or to $=$) 
create a new positive variable and add it with coefficient 1 to the left part of the inequality.
- From $=$ to $\leq$ (or to $\geq$)
put two versions one with $\leq$ and the other with $\geq$, multiply the last one by -1 .
- From x unrestricted to non-negative variables,
create two new variables x^+ and x^- , both non negative, replace x by $x^+ - x^-$.

Linear programming: standard formulation

LP standard form

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \geq b \\ & x \geq 0\end{array}$$

Where

- $x = (x_1, \dots, x_n)$, $c = (c_1, \dots, c_n)$.
- $b^T = (b_1, \dots, b_m)$
- A is a $n \times m$ matrix.

Linear programming: problem

Given

- $c = (c_1, \dots, c_n)$,
- $b^T = (b_1, \dots, b_m)$,
- and a $n \times m$ matrix A .

find $x = (x_1, \dots, x_n) \geq 0$, so that

- $Ax \geq b$ and $c^T x$ is minimized.

Linear programming: algorithms

We can solve Linear Programming in polynomial time



- Simplex method: Dantzig in 1947 (exponential time Klee and Minty 1972)
- Ellipsoid method: Khachiyan 1979 ($O(n^6)$)
- Interior-point method: Karmarkar 1984 ($O(n^3)$)
- Most used algorithm is still Simplex (fast on average).
- Many commercial LP solvers CPLEX and open source Gurobi

Integer programming

- An **integer programming (IP)** problem is a linear programming problem with the additional restriction that the **values of the variables must be integers**.
- A **mixed integer programming (MIP)** problem is a linear programming problem with the additional restriction that, the **values of some variables must be integers**.
- Many NPO problems can be easily expressed as IP or MIP problems
- IP is NP-hard

Max SAT as integer program

- **Max Sat:** Input a set of m clauses on n variables, find an assignment that maximizes the number of satisfied clauses.
- For a clause j , the set of variables that appear in C_j
 - positive is $P(j)$
 - negative is $N(j)$
- We consider $n + m$ integer variables,
 - $x_1, \dots, x_n$, one per each variable
 - $y_1, \dots, y_m$, one per each clause

The variables will be restricted to have values in $\{0, 1\}$

This is a simplification of saying that they must hold integer values and that all of them are ≤ 1 .

Max SAT as integer program

Max SAT-IP

$$\begin{array}{ll}
 \max & \sum_{j=1}^m y_j \\
 \text{s.t.} & \sum_{i \in P(j)} x_i + \sum_{i \in N(j)} (1 - x_i) \geq y_j \quad 1 \leq j \leq m \\
 & y_j \in \{0, 1\} \quad 1 \leq j \leq m \\
 & x_i \in \{0, 1\} \quad 1 \leq i \leq n
 \end{array}$$

The size of the IP is polynomial in the size of the Max SAT, so the transformation is a polynomial Turing reduction from Max SAT to IP.

Vertex cover as integer program

VC

Given a graph $G = (V, E)$ we want to find a set $S \subset V$ with minimum cardinality, so that every edge in G has at least one end point in S .

VC-IP

$$\begin{array}{ll} \min & \sum_{i=1}^n x_i \\ \text{s.t.} & x_i + x_j \geq 1 \quad \text{for all } (i, j) \in E \\ & x_i \in \{0, 1\} \quad \text{for all } i \in V \end{array}$$

Weighted Vertex cover as integer program

WVC

Given a graph $G = (V, A)$ with weights w associated to the vertices, we want to find a set $S \subset V$ with minimum weight, so that every edge in G has at least one end point in S .

VC-IP

$$\begin{array}{ll} \min & \sum_{i=1}^n w_i x_i \\ \text{s.t.} & x_i + x_j \geq 1 \quad \text{for all } (i, j) \in E \\ & x_i \in \{0, 1\} \quad \text{for all } i \in V \end{array}$$

Exercise

Try to write a LP or IP formulation for the problems

- Min Weighted Matching
- Set cover
- Max Flow

- 1 LP and IP
- 2 Relax and round**
- 3 LP Duality

Relaxation and rounding

- Many real-life problems can be modeled as Integer Linear Programs (IP).
- The IP can be relaxed to a linear program (LP) by eliminating the integrity constraints. (no more constraints LP)
- By doing so the optimum cost can only improve, i.e.,
opt of LP is better than opt of IP. (better solutions) (unrestricted minimum)
- We can solve the LP in polynomial time.
- The LP optimal solution might not be integral when possible,
 ↗ transform it to get a feasible integer solution not far from opt of IP.
 (rounding, param enter)

Vertex cover

VC

Given a graph $G = (V, A)$ we want to find a set $S \subset V$ with minimum cardinality, so that every edge in G has at least one end point in S .

el relaxar

VC-IP

$$\begin{aligned} \min \quad & \sum_{i=1}^n x_i \\ \text{s.t.} \quad & x_i + x_j \geq 1 \quad \text{for all } (i, j) \in E \\ & x_i \in \{0, 1\} \quad \text{for all } i \in V \end{aligned}$$

(1 variable per cada node,
val 1 si hi és a S i 0 si no)

VC-LP

$$\begin{aligned} \min \quad & \sum_{i=1}^n x_i \\ \text{s.t.} \quad & x_i + x_j \geq 1 \quad \text{for all } (i, j) \in E \\ & x_i \geq 0 \quad \text{for all } i \in V \end{aligned}$$

(no cal parer $x_i \leq 1$, sempre es compleix)

Vertex cover: another approximation algorithm

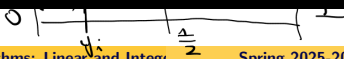
VC-LP has an optimal solution x^* such that $x_i \in \{0, 1, 1/2\}$.
Furthermore, such a solution can be computed in polynomial time.

Let y be an optimal solution s.t. not all its coordinates are in $\{0, 1, 1/2\}$.

Set $\epsilon = \min_{y_i \notin \{0, 1, 1/2\}} \{y_i, |y_i - 1/2|, 1 - y_i\}$. Consider $\epsilon \equiv$ a quin e d i c i e n t e p e r o c a d o y_i

$$\begin{array}{l}
 \text{(le para a lla)} \\
 y'_i = \begin{cases} y_i - \epsilon & 0 < y_i < 1/2 \\ y_i + \epsilon & 1/2 < y_i < 1 \\ y_i & \text{otherwise} \end{cases}
 \end{array}
 \quad
 \begin{array}{l}
 \text{(le para a lla)} \\
 y''_i = \begin{cases} y_i + \epsilon & 0 < y_i < 1/2 \\ y_i - \epsilon & 1/2 < y_i < 1 \\ y_i & \text{otherwise} \end{cases}
 \end{array}$$

$\sum y_i = (\sum y'_i + \sum y''_i)/2$, so both are optimal solutions. One of them has more $\{0, 1, 1/2\}$ coordinates than y . ϵ repete fins a que um h e tot $\{0, 1, \frac{1}{2}\}$ ☺



Vertex cover

function RELAX+ROUND VC(G)

Construct the LP-VC associated G

Let y be an optimal relaxed solution (of the LP instance)

Using the previous lemma, construct an optimal relaxed solution y' such that $y'_i \in \{0, 1, 1/2\}$

Let x defined as $x_i = 0$ if $y'_i = 0$, $x_i = 1$ otherwise.

return (x)

RELAX+ROUND VC

- runs in polynomial time
- x defines a vertex cover
- $\sum_{i=1}^n x_i \leq 2 \sum_{i=1}^n y'_i \leq \underline{\underline{2\text{opt}}}$
- is a 2-approximation for VC.

• $\forall e = (i, j) \quad y'_i + y'_j \geq 1$, e garantir o tamanho m' !

Weighted vertex cover: Relax+Round approximation

LP WVC

$$\begin{aligned}
 \min \quad & \sum_{i=1}^n w_i x_i \\
 \text{s.t.} \quad & x_i + x_j \geq 1 \quad \text{for all } (i, j) \in E \\
 & x_i \geq 0 \quad \text{for all } i \in V
 \end{aligned}$$

function WVC(G, c)

Construct the LP WVC, I

$y = LP.solve(I)$

for $i = 1, \dots, n$ **do**

if $y_i < 1/2$ **then**

$x_i = 0$

else ($\geq \frac{1}{2}$)

$x_i = 1$

} rounding

return (x)

RELAX+ROUND WVC

- runs in polynomial time
- x defines a vertex cover
- $\sum_{i=1}^n w_i x_i \leq 2 \sum_{i=1}^n w_i y_i \leq 2 \text{opt}$
- is a 2-approximation for WVC.

Minimum 2-Satisfiability

MIN 2-SAT

Given a Boolean formula in 2-CNF, determine whether it is satisfiable and, in such a case, find a satisfying assignment with minimum number of true variables.

- 2-SAT can be solved in polynomial time. ☺
- MIN 2-SAT is NP-hard. ☹
- MIN 2-SAT IP formulation?

Minimum 2-Satisfiability: IP formulation

Suppose that F has n variables $x_1, \dots, x_n$ and m clauses with 2 literals per clause

IP Min 2-SAT

$$\begin{aligned}
 & \min \quad \sum_{i=1}^n x_i \\
 \text{s.t.} \quad & x_i + x_j \geq 1 \quad \text{for all clauses } (x_i \vee x_j) \in F \quad (+, +) \\
 & (1 - x_i) + x_j \geq 1 \quad \text{for all clauses } (\bar{x}_i \vee x_j) \in F \quad (-, +) \\
 & (1 - x_i) + (1 - x_j) \geq 1 \quad \text{for all clauses } (\bar{x}_i \vee \bar{x}_j) \in F \quad (-, -) \\
 & x_i \in \{0, 1\} \quad 1 \leq i \leq n
 \end{aligned}$$

LP Min 2-SAT is obtained by replacing $x_i \in \{0, 1\}$ by $x_i \geq 0$.

Minimum 2-Satisfiability: LP relaxation

LP Min 2-SAT

$$\begin{array}{ll}
 \min & \sum_{i=1}^n x_i \\
 \text{s.t.} & x_i + x_j \geq 1 \quad \text{for all clauses } (x_i \vee x_j) \in F \\
 & (1 - x_i) + x_j \geq 1 \quad \text{for all clauses } (\bar{x}_i \vee x_j) \in F \\
 & (1 - x_i) + (1 - x_j) \geq 1 \quad \text{for all clauses } (\bar{x}_i \vee \bar{x}_j) \in F \\
 & x_i \geq 0 \quad 1 \leq i \leq n
 \end{array}$$

- Let y be an optimal solution to LP Min 2-SAT.
- Can we use the same rounding scheme as for [WVC](#)?
- Setting $x_i = 1$ if $y_i > 1/2$ and $x_i = 0$ if $y_i < 1/2$ is safe, all clauses with at least one literal with value $> 1/2$ will be satisfied.
- When $y_i = 1/2$?

Minimum 2-Satisfiability: LP relaxation

- Let y be an optimal solution to IP Min 2-SAT.
- What to do when $y_i = 1/2$? 1? 0?
- If F contains the clauses $(x_i \vee x_j)$ and $(\bar{x}_i \vee \bar{x}_j)$ and $y_i = y_j = 1/2$, neither $x_i = x_j = 1$ nor $x_i = x_j = 0$ satisfy the formula.
- F_1 = clauses whose two variables have y value $= 1/2$. (we're at a local minimum)
- Rounding those values to 1 or 0 would keep the approximation ratio to 2, provided the constructed solution x to **MIN 2-SAT** is still a satisfying assignment.
- Any satisfying assignment for the clauses in F_1 and get a 2-approximation 😊

Minimum 2-Satisfiability: Relax+Round approximation

function RELAX+ROUND MIN 2-SAT(F)

if F is not satisfiable **then return** false

 Construct the LP Min 2-SAT, I

$y = LP.solve(I)$

for $i = 1, \dots, n$ **do**

if $y'_i < 1/2$ **then** $x_i = 0$

if $y'_i > 1/2$ **then** $x_i = 1$

$F_1 =$ clauses with both y values $= 1/2$.

 Let $J = \{j \mid x_j \in F_1\}$

for $i=1, \dots, n$ **do**

if $y_i = 1/2$ and $i \notin J$ **then** $x_i = 1$ | *no else or double conflict, do not force a 1*

 Complete x with a satisfying assignment for F_1

return (x)

Minimum 2-Satisfiability: Relax+Round approximation

RELAX+ROUND MIN 2-SAT is a 2-approximation for **MIN 2-SAT**.

Max Satisfiability

MAX SAT

Given a Boolean formula in CNF and weights for each clause, find a Boolean assignment to maximize the weight of the satisfied clauses.

Suppose that F has n variables $x_1, \dots, x_n$ and m clauses $C_1, \dots, C_m$.

IP Max SAT

$$\begin{aligned} \max \quad & \sum_{j=1}^m w_j z_j \\ \text{s.t.} \quad & \sum_{x_i \in C_j} y_i + \sum_{\bar{x}_i \in C_j} (1 - y_i) \geq z_j \quad j = 1, \dots, m \\ & y_i \in \{0, 1\} \quad 1 \leq i \leq n \\ & z_j \in \{0, 1\} \quad 1 \leq j \leq m \end{aligned}$$

LP Max SAT is obtaining replacing $a \in \{0, 1\}$ by $0 \leq a \leq 1$.

quantified variable

Max Satisfiability: Relax+RRound

```

function RELAX+RRound( $F$ )
    Construct the LP Max SAT,  $I$ 
     $(y, z) = LP.solve(I)$ 
    for  $i=1, \dots, n$  do
        Set  $x_i = 1$  with probability  $y_i$ 
    return  $(x)$ 

```

- The optimal LP solution is used as an indicator of the probability that the variable has to be set to 1.
- The performance of a randomized algorithm is the expected number of satisfiable clause.
- This expectation has to be compared with opt.

Max Satisfiability: Relax+RRound

- Let (y^*, z^*) be an optimal solution of LP Max SAT
- Let Z_j be the indicator random variable for the event that clause C_j is satisfied.
- Assume that C_j has k -literals and that ℓ of them are negated variables.

For any $1 \leq j \leq m$, $E[Z_j] \geq z_j^*(1 - 1/e)$.

Recall $(a_1 \dots a_k)^{1/k} \leq (a_1 + \dots + a_k)/k$ or equivalently

$$(a_1 \dots a_k) \leq ((a_1 + \dots + a_k)/k)^k$$

(multi)

(Summation)

Max Satisfiability: Relax+RRound

Z_j is an indicator random variable, and so

$$E[Z_j] = \Pr[Z_j = 1] = 1 - \Pr[Z_j = 0]$$

(x_i from 0) ($\bar{x}_i$ from 1)

⤵ Regio anterior

$$\Pr[Z_j = 0] = \prod_{x_i \in C_j} (1 - y_i^*) \cdot \prod_{\bar{x}_i \in C_j} y_i^* \leq \left(\frac{(k - \ell) - \sum_{x_i \in C_j} y_i^* + \sum_{\bar{x}_i \in C_j} y_i^*}{k} \right)^k$$

$$\leq \left(\frac{(k - \sum_{x_i \in C_j} y_i^* - \sum_{\bar{x}_i \in C_j} (1 - y_i^*))}{k} \right)^k \leq \left(\frac{(k - z_j^*)}{k} \right)^k \leq \left(1 - \frac{z_j^*}{k} \right)^k$$

$$E[Z_j] \geq 1 - \left(1 - \frac{z_j^*}{k} \right)^k \geq z_j^* \left(1 - \frac{1}{k} \right)^k \geq z_j^* (1 - 1/e)$$



Max Satisfiability: Relax+RRound approximation

RELAX+RRound is a $e/(e-1)$ -approximation for **MAX SAT**.

- Let (y^*, z^*) be an optimal solution of LP Max SAT
- Let Z_j be the indicator r.v.a for clause C_j is satisfied.
- Let W be the r.v. weight of satisfied clauses: $W = \sum_{j=1}^m w_j Z_j$.
- $E[W] = \sum_{j=1}^m w_j E[Z_j] \geq (1 - 1/e) \sum_{j=1}^m w_j z_j^* \geq (1 - 1/e) \text{opt}$



Max Satisfiability: RandAssign

```

function RANDASSIGN( $F$ )
  for  $i=1, \dots, n$  do
    Set  $x_i = 1$  with probability  $1/2$ 
  return ( $x$ )

```

RANDASSIGN is a 2-approximation for MAX SAT.

$$E[W] = \sum_{j=1}^m w_j E[Z_j] = \sum_{j=1}^m w_j \left(1 - \left(\frac{1}{2}\right)^{k_j}\right) \geq \frac{1}{2} \sum_{j=1}^m w_j \geq \frac{1}{2} \text{opt.} \quad \text{😊}$$

We move from $r = 2$ (RANDASSIGN) to $r = 1.581977$ (RELAX+RRound).

Max Satisfiability: Best2

```
function BEST2( $F$ )  
   $x_1, W_1 = \text{RANDASSIGN}(F)$   
   $x_2, W_2 = \text{RELAX+RRROUND}(F)$   
  if  $W_1 \geq W_2$  then  
    return ( $x_1$ )  
  else  
    return ( $x_2$ )
```

BEST2 is a $4/3$ (1.33333)-approximation for **MAX SAT**.

Max Satisfiability: Best2

- $E[W] = E[\max\{W_1, W_2\}] \geq E[(W_1 + W_2)/2]$.

$$\begin{aligned}
 E[W] &\geq \sum_{j=1}^m w_j \left[\frac{1}{2} \left(1 - \left(\frac{1}{2} \right)^{k_j} \right) + \frac{1}{2} z_j^* \left(1 - \left(\frac{1}{k_j} \right)^{k_j} \right) \right] \\
 &\geq \sum_{j=1}^m w_j \frac{3}{4} z_j^* \geq \frac{3}{4} \sum_{j=1}^m w_j z_j^* \geq \frac{3}{4} \text{opt.}
 \end{aligned}$$

Max Satisfiability: Best2

- Is $\left[\frac{1}{2} \left(1 - \left(\frac{1}{2} \right)^{k_j} \right) + \frac{1}{2} z_j^* \left(1 - \left(\frac{1}{k_j} \right)^{k_j} \right) \right] \geq \frac{3}{4} z_j^*$?
- $k_j = 1$: $\frac{1}{2} \frac{1}{2} + \frac{1}{2} z_j^* \geq \frac{3}{4} z_j^*$.
- $k_j = 2$: $\frac{1}{2} \frac{3}{4} + \frac{1}{2} \frac{3}{4} z_j^* \geq \frac{3}{4} z_j^*$.
- $k_j \geq 3$: the minimum possible of each term is

$$\frac{1}{2} \frac{7}{8} + \frac{1}{2} \left(1 - \frac{1}{e} \right) z_j^* \geq \frac{3}{4} z_j^*$$



- 1 LP and IP
- 2 Relax and round
- 3 LP Duality**